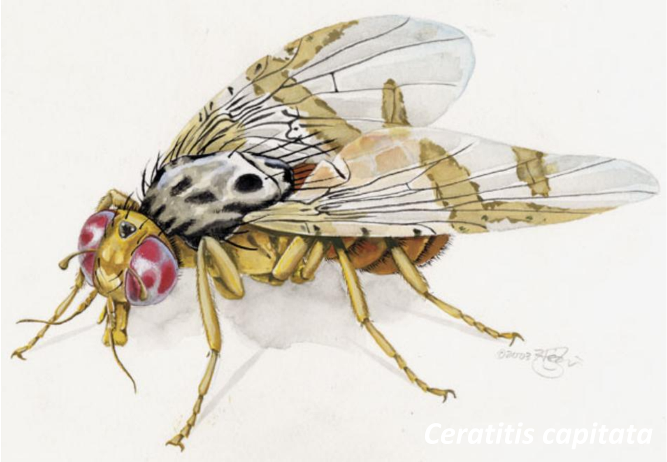
STOPTIGRE project: community engagement using art and science transdisciplinary approach

Sex determination in insects

SEX DETERMINATION

Maleness-on-the-Y (*MoY*) orchestrates male sex determination in major agricultural fruit fly pests

Angela Meccariello[1*], Marco Salvemini[1*], Pasquale Primo[1], Brantley Hall[2], Panagiota Koskinioti[3,4], Martina Dalíková[5,6], Andrea Gravina[1], Michela Anna Gucciardino[1], Federica Forlenza[1], Maria-Eleni Gregoriou[4], Domenica Ippolito[1], Simona Maria Monti[7], Valeria Petrella[1], Maryanna Martina Perrotta[1], Stephan Schmeing[8], Alessia Ruggiero[7], Francesca Scolari[9], Ennio Giordano[1], Konstantina T. Tsoumani[4], František Marec[5], Nikolai Windbichler[10], Kallare P. Arunkumar[11], Kostas Bourtzis[3], Kostas D. Mathiopoulos[4], Jiannis Ragoussis[12], Luigi Vitagliano[7], Zhijian Tu[2], Philippos Aris Papathanos[13,14]†, Mark D. Robinson[8]†, Giuseppe Saccone[1]†



Meccariello *et al.*, *Science* **365**, 1457–1460 (2019)   27 September 2019

Salvemini *et al. BMC Evolutionary Biology* 2011, **11**:41
http://www.biomedcentral.com/1471-2148/11/41

BMC Evolutionary Biology

RESEARCH ARTICLE   Open Access

Genomic organization and splicing evolution of the *doublesex* gene, a *Drosophila* regulator of sexual differentiation, in the dengue and yellow fever mosquito *Aedes aegypti*

Marco Salvemini[1†], Umberto Mauro[1,2†], Fabrizio Lombardo[3], Andreina Milano[1], Vincenzo Zazzaro[1], Bruno Arcà[3,4], Lino C Polito[1,5], Giuseppe Saccone[1*]

OPEN ACCESS Freely available online   PLOS | ONE

The Orthologue of the Fruitfly Sex Behaviour Gene *Fruitless* in the Mosquito *Aedes aegypti*: Evolution of Genomic Organisation and Alternative Splicing

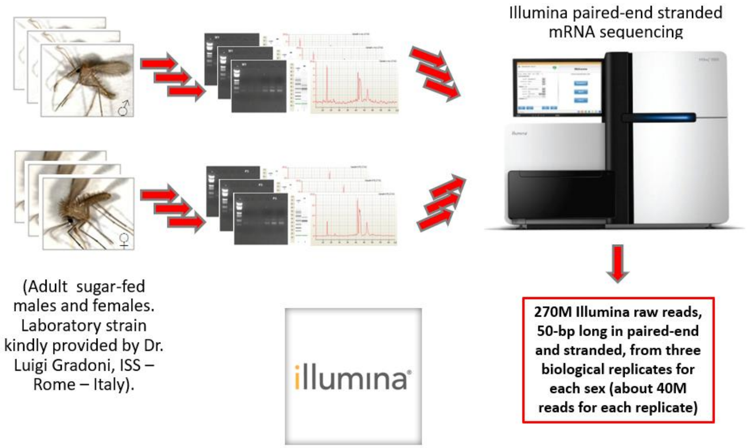
Marco Salvemini[1*], Rocco D'Amato[1], Valeria Petrella[1], Serena Aceto[1], Derric Nimmo[2], Marco Neira[2], Luke Alphey[2,3], Lino C. Polito[1], Giuseppe Saccone[1]

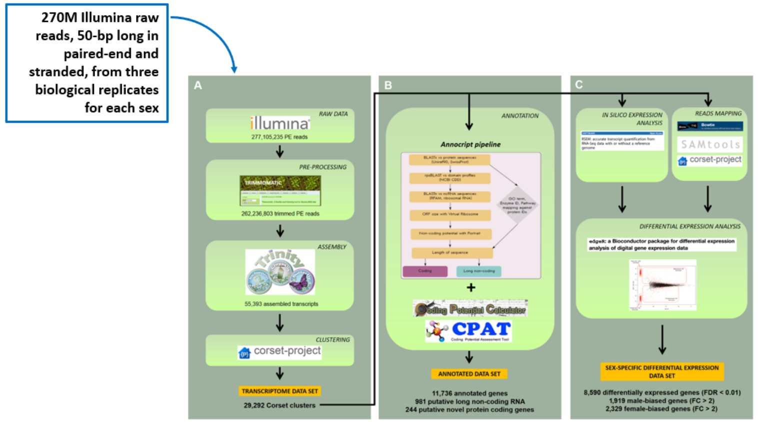
1 Department of Biological Sciences – Section of Genetics and Molecular Biology, University of Naples "Federico II", Naples, Italy, **2** Oxitec Limited, Oxford, United Kingdom, **3** Department of Zoology, University of Oxford, Oxford, United Kingdom

Department of Biology University of Naples Federico II

2014

Sex determination in insects



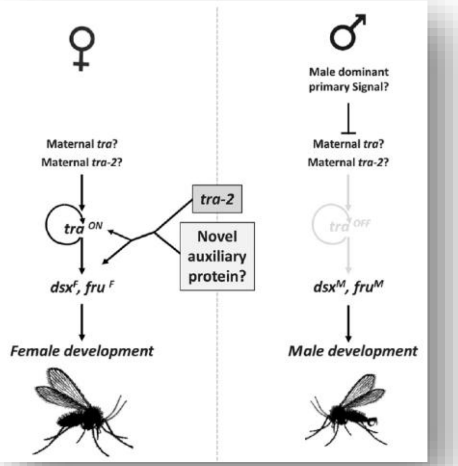


Petrella *et al. BMC Genomics* (2015) 16:847
DOI 10.1186/s12864-015-2088-x
BMC Genomics

RESEARCH ARTICLE **Open Access**

De novo assembly and sex-specific transcriptome profiling in the sand fly *Phlebotomus perniciosus* (Diptera, Phlebotominae), a major Old World vector of *Leishmania infantum*

V. Petrella[1], S. Aceto[1], F. Musacchia[2], V. Colonna[3], M. Robinson[4,5], V. Benes[6], G. Cicotti[7], G. Bongiorno[8], L. Gradoni[8], P. Volf[9] and M. Salvemini[1*]



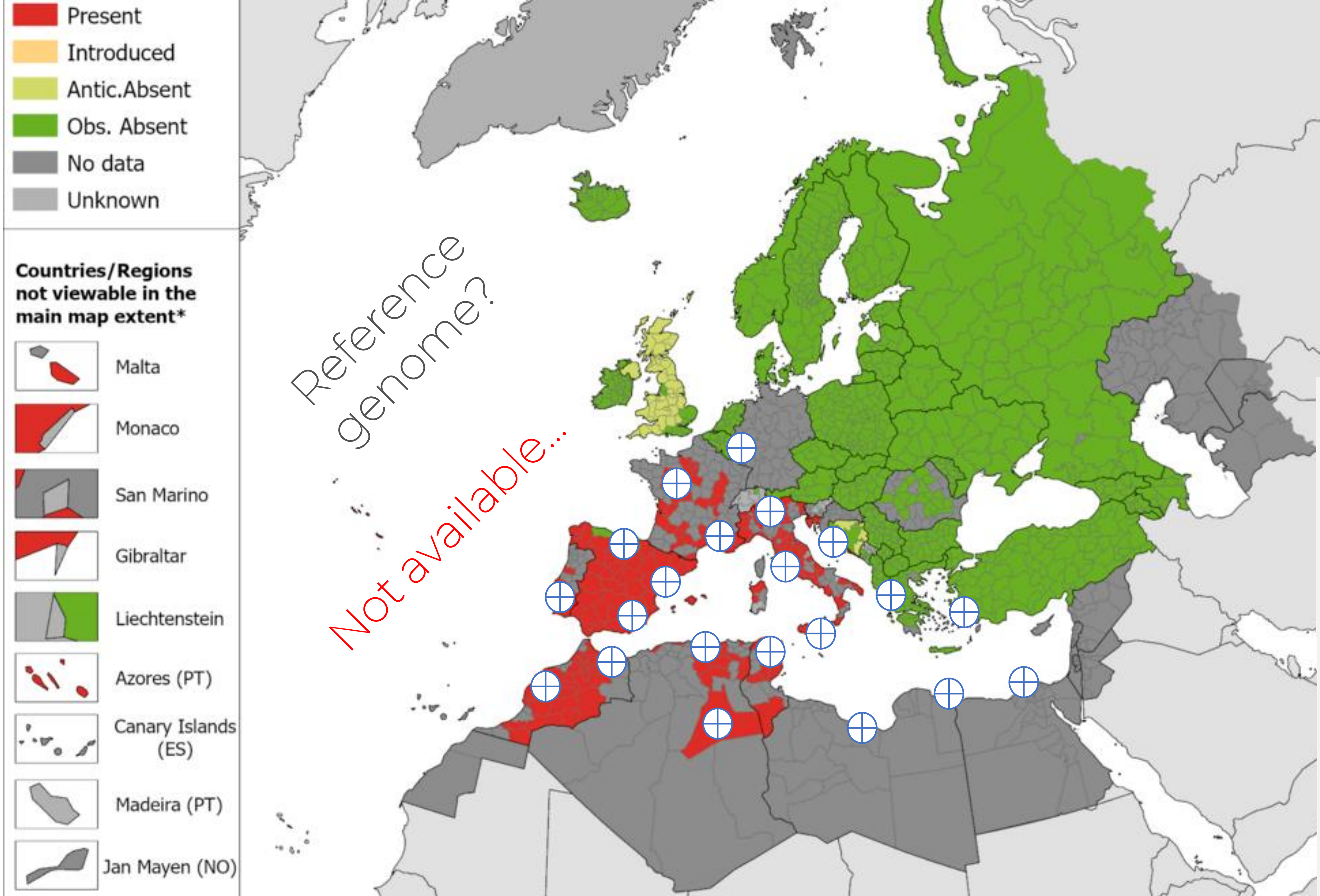
Petrella *et al. BMC Genomics* (2019) 20:522
https://doi.org/10.1186/s12864-019-5898-4
BMC Genomics

RESEARCH ARTICLE **Open Access**

Identification of sex determination genes and their evolution in Phlebotominae sand flies (Diptera, Nematocera)

Valeria Petrella[1], Serena Aceto[1], Vincenza Colonna[2], Giuseppe Saccone[1], Remo Sanges[3,4], Nikola Polanska[5], Petr Volf[5], Luigi Gradoni[6], Gioia Bongiorno[6] and Marco Salvemini[1*]

First population genomics analysis of *Phlebotomus perniciosus*



Serena Aceto

Vincenza Colonna

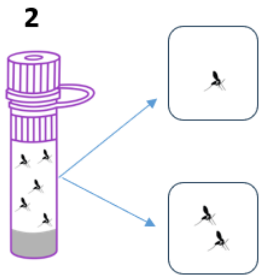
Primary assembly 2026

The sandfly *P. perniciosus* genome project: 2nd attempt with 1 individual

90ng di HMW gDNA extracted. High quality Pac-Bio genome sequencing for P. perniciosus resulting in ~67x HiFi coverage (12,0 Gb HiFi reads with an average length of 9,000 bp).

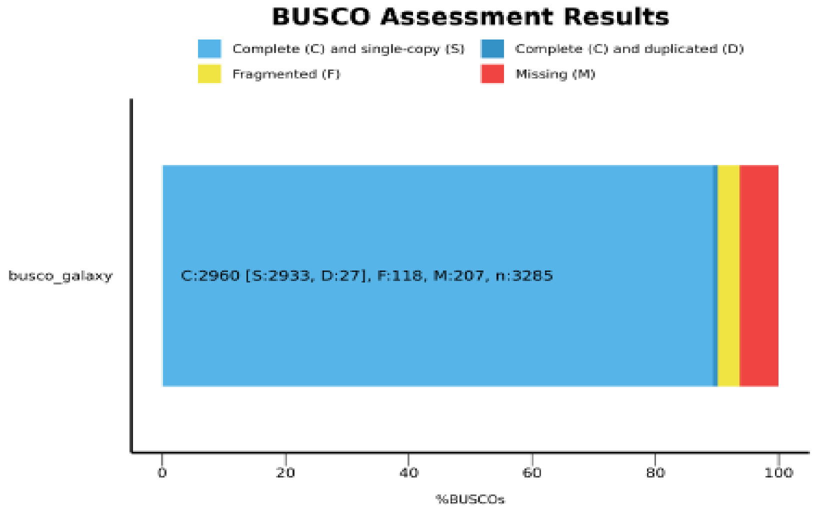
1 individual

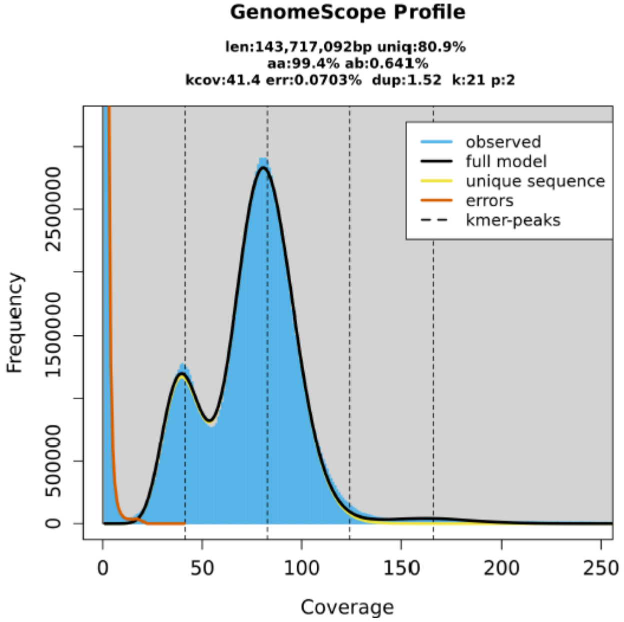
Title: SFPS048-11 Lateral
Record licence: http://creativeco.../licenses/by/4.0/
References: http://bins.bolds...cessid=SFPS048-11
Identifier: http://www.boldsy...48+1330027796.JPG
Suggested attribution: "SFPS048-11 Lateral" - BOLD:AAY5165 (cf. *Phlebotomus perniciosus*) (licensed under http://creativecommons.org/licenses/by/4.0/)



HiFi PacBio sequencing

HiC Arima Sequencing

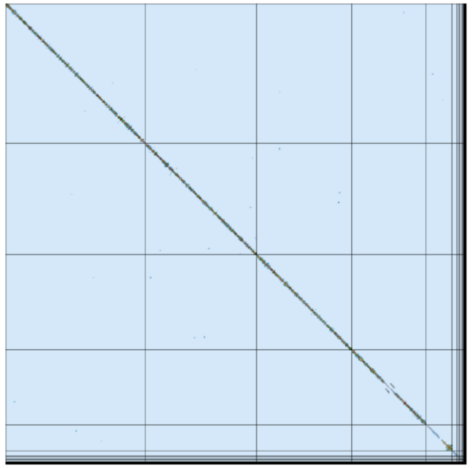




S1 Primary

Expected genome size	143,801,699
# scaffolds	21
Total scaffold length	150,235,544
Average scaffold length	7,154,073.52
Scaffold N50	36,611,036
Scaffold auN	33,592,830.57
Scaffold L50	2
Scaffold NG50	36,611,036
Scaffold auNG	35,095,810.48
Scaffold LG50	2
Largest scaffold	45,557,476
Smallest scaffold	11,360

N50 - L50



HiC scaffolding

In collaboration with:

Giulio Formenti

VGL

The Rockefeller University 1901
PRO BONO HUMANI GENERIS SCIENTIA

2016

https://stoptigre.evosexdevo.eu/

Citizen-science
Research
Third mission

Mosquitoes are considered the world's deadliest animal species

World's deadliest Animals

Number of humans killed by animals, 2015

100 Elephant

8,000 kissing bug

4,400 Fresh water snail

1,600 Tapeworm

24,200 Sandfly

830,000 Mosquitoes

500 Hippopotamus

3,500 Scorpion

6 Shark

17,400 Dog

40 Jellyfish

60 Bee

60,000 Snake

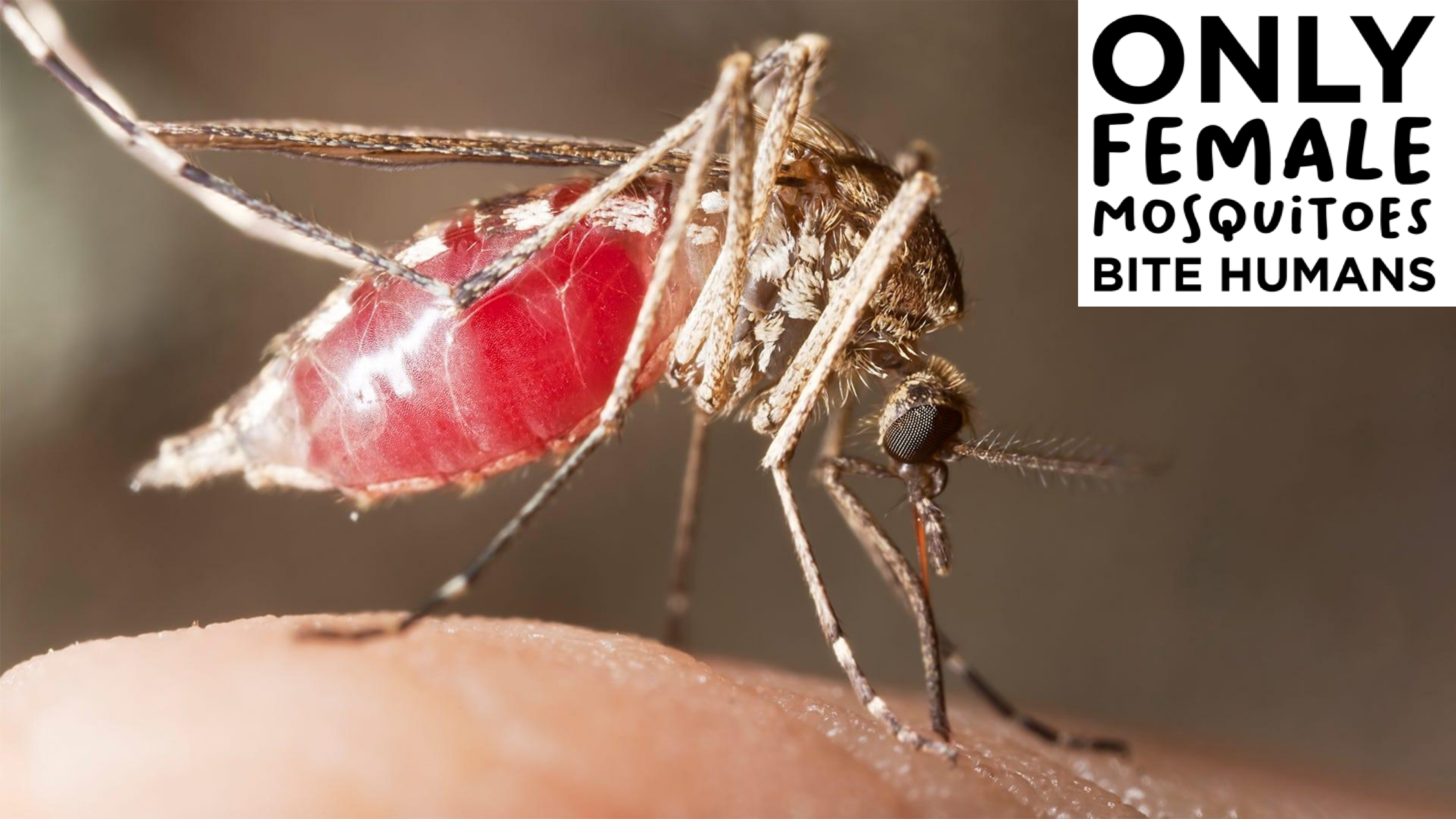
3,500 Tsetse fly

SOURCES: IHME, WHO, CrocBITE, FAO, Norvegian Institute for Nature Research, International Shark Attack File, National Geographic, PBS, National Science Foundation, CDC, WWF, French Institute of Research for Development, Wilderness & Environmental Mediche, Nature. All calculations have wide error margins.

Fighting insect pests with pesticides: a blessing and curse approach

Food and Agriculture Organization
of the United Nations

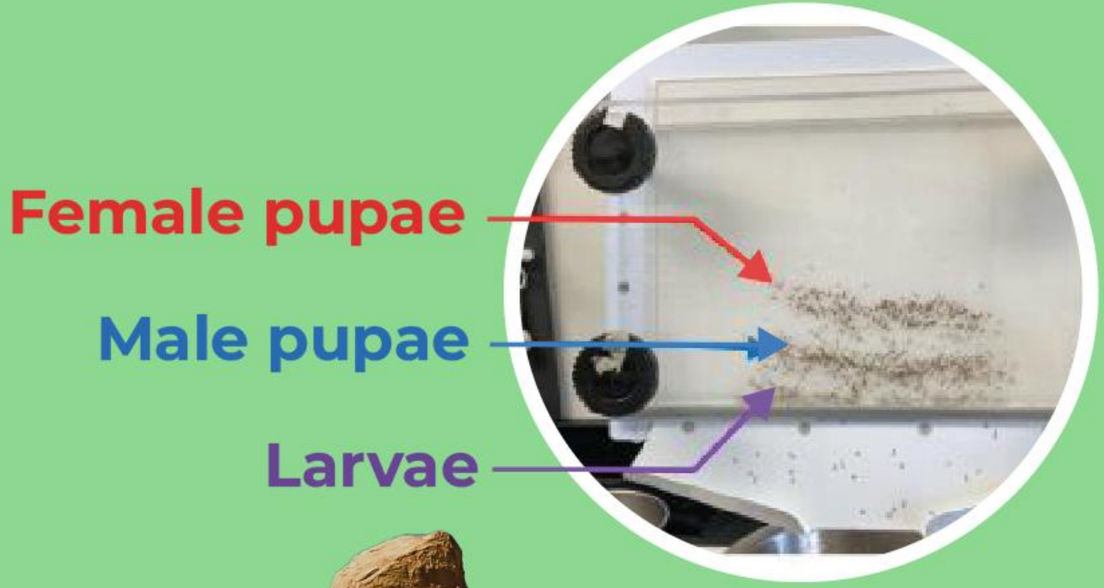
IAEA
International Atomic Energy Agency
Atoms for Peace


ONLY
FEMALE
MOSQUITOES
BITE HUMANS

MAJOR SHORTCOMINGS THAT LIMIT SIT/IIT APPLICATION TO MOSQUITOES ON A LARGE SCALE

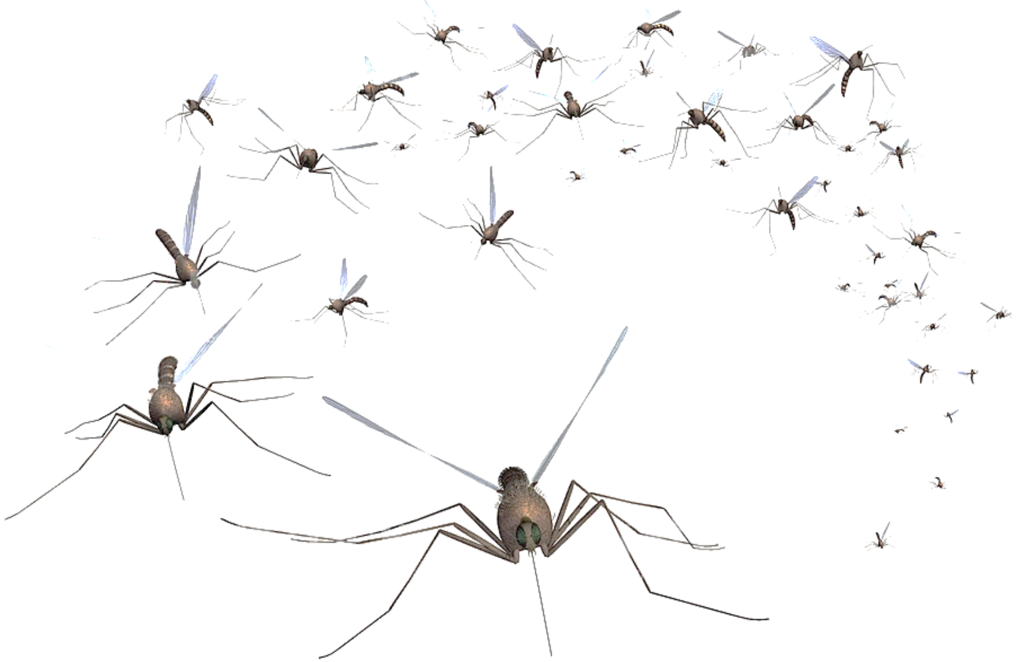
1) Absence of an efficient sex separation technology

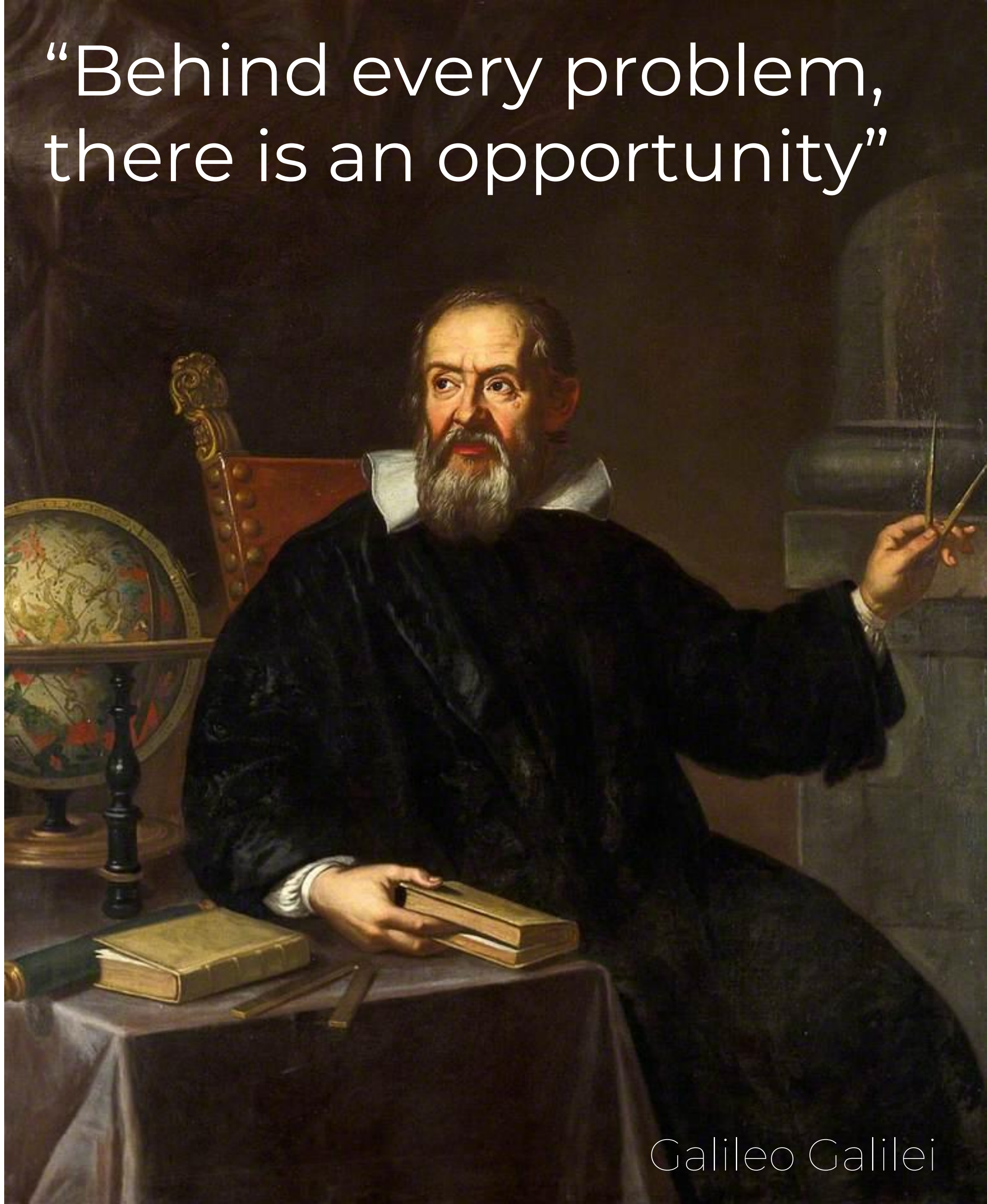
No 100% efficient sex sorting system available to date at an industrial scale.



MAJOR SHORTCOMINGS THAT LIMIT SIT/IIT APPLICATION TO MOSQUITOES ON A LARGE SCALE

2) Without effective community participation, eco-friendly control campaigns could not be achieved




"Behind every problem,
there is an opportunity"
Galileo Galilei

Ecofriendly tiger mosquito suppression on Procida Island through community participation

Procida Science Island 2016

Raimondo Ambrosino, Procida **Mayor**

Rossella Lauro, counselor

Giuseppe Saccone

Titta Lubrano, **counselor**

Vincenza Colonna IBG-CNR

2016

Promotion of scientific culture on the Island by organizing scientific events

2019

2023

COMMUNITY ENGAGEMENT WITH THE HELP OF THE PROCIDA ADMINISTRATION

2016

Temporal dynamics analysis over one year using ovitraps involving 12 volunteers

RAIMONDO AMBROSINO

PROCIDA MAYOR

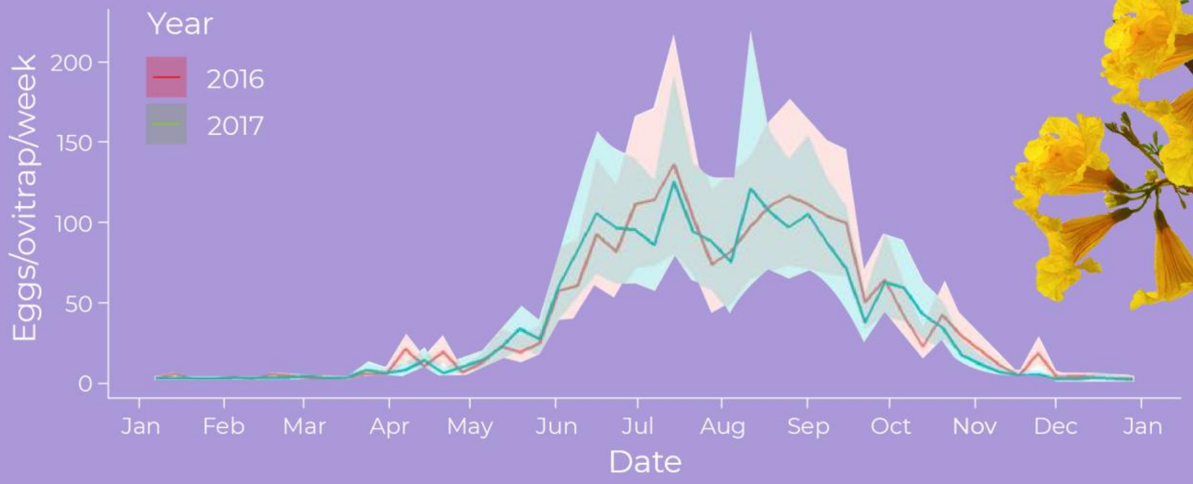
ROSSELLA LAURO

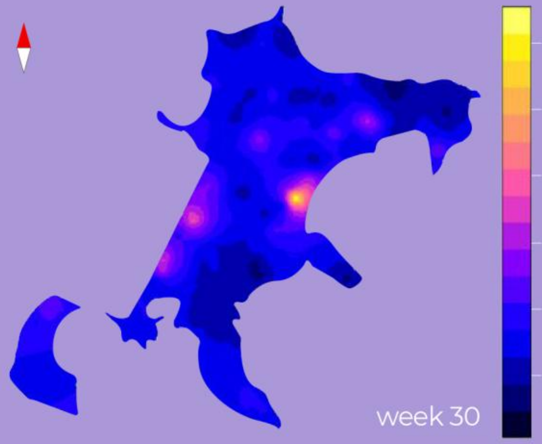
MUNICIPAL CONSELOR

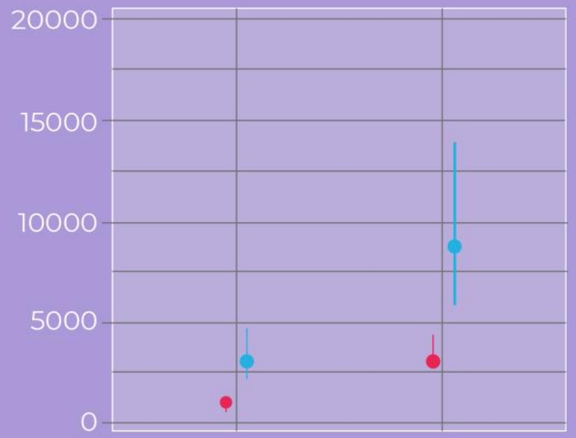
2016

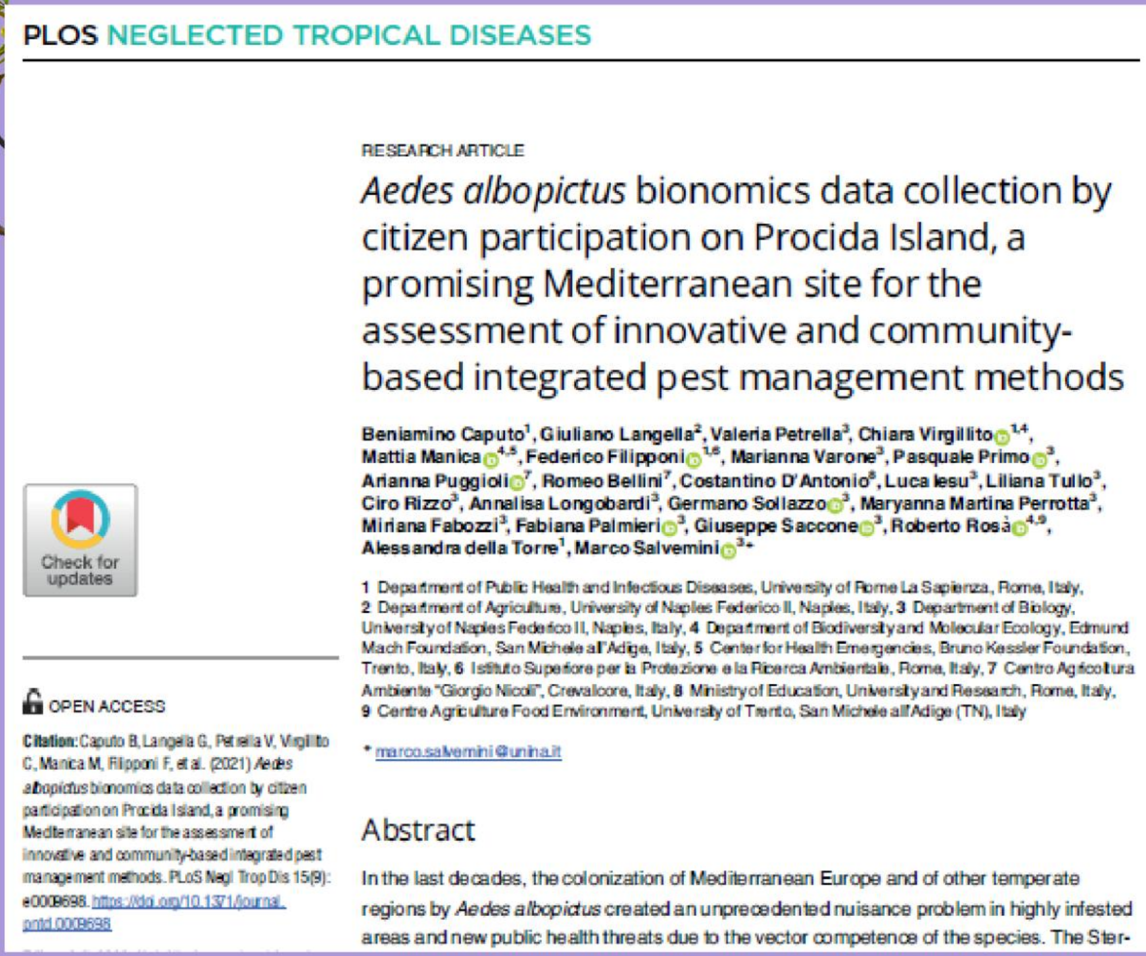
CITIZEN-SCIENCE: CITIZENS BECOME PROTAGONISTS!
SPATIAL DYNAMICS ANALYSIS - INVOLVING 80 FAMILIES - 2017

With the active participation of Citizen, we collected baseline data about *AEDES ALBOPICTUS* bionomics on Procida Island









PLOS NEGLECTED TROPICAL DISEASES

RESEARCH ARTICLE

Aedes albopictus bionomics data collection by citizen participation on Procida Island, a promising Mediterranean site for the assessment of innovative and community-based integrated pest management methods

Beniamino Caputo[1], Giuliano Langella[2], Valeria Petrella[3], Chiara Virgillito[1,4], Mattia Manica[4,5], Federico Filipponi[1,6], Marianna Varone[3], Pasquale Primo[3], Arianna Puggioli[7], Romeo Bellini[7], Costantino D'Antonio[8], Luca Iesu[3], Liliana Tullo[3], Ciro Rizzo[3], Annalisa Longobardi[3], Germano Sollazzo[3], Maryanna Martina Perrotta[3], Miriana Fabozzi[3], Fabiana Palmieri[3], Giuseppe Saccone[3], Roberto Rosà[4,9], Alessandra della Torre[1], Marco Salvemini[3*]

1 Department of Public Health and Infectious Diseases, University of Rome La Sapienza, Rome, Italy, 2 Department of Agriculture, University of Naples Federico II, Naples, Italy, 3 Department of Biology, University of Naples Federico II, Naples, Italy, 4 Department of Biodiversity and Molecular Ecology, Edmund Mach Foundation, San Michele all'Adige, Italy, 5 Center for Health Emergencies, Bruno Kessler Foundation, Trento, Italy, 6 Istituto Superiore per la Protezione e la Ricerca Ambientale, Rome, Italy, 7 Centro Agricoltura Ambiente "Giorgio Nicoli", Crevalcore, Italy, 8 Ministry of Education, University and Research, Rome, Italy, 9 Centre Agriculture Food Environment, University of Trento, San Michele all'Adige (TN), Italy

* marco.salvemini@unina.it

Check for updates

OPEN ACCESS

Citation: Caputo B, Langella G, Petrella V, Virgillito C, Manica M, Filipponi F, et al. (2021) *Aedes albopictus* bionomics data collection by citizen participation on Procida Island, a promising Mediterranean site for the assessment of innovative and community-based integrated pest management methods. PLoS Negl Trop Dis 15(9): e0009698. https://doi.org/10.1371/journal.pntd.0009698

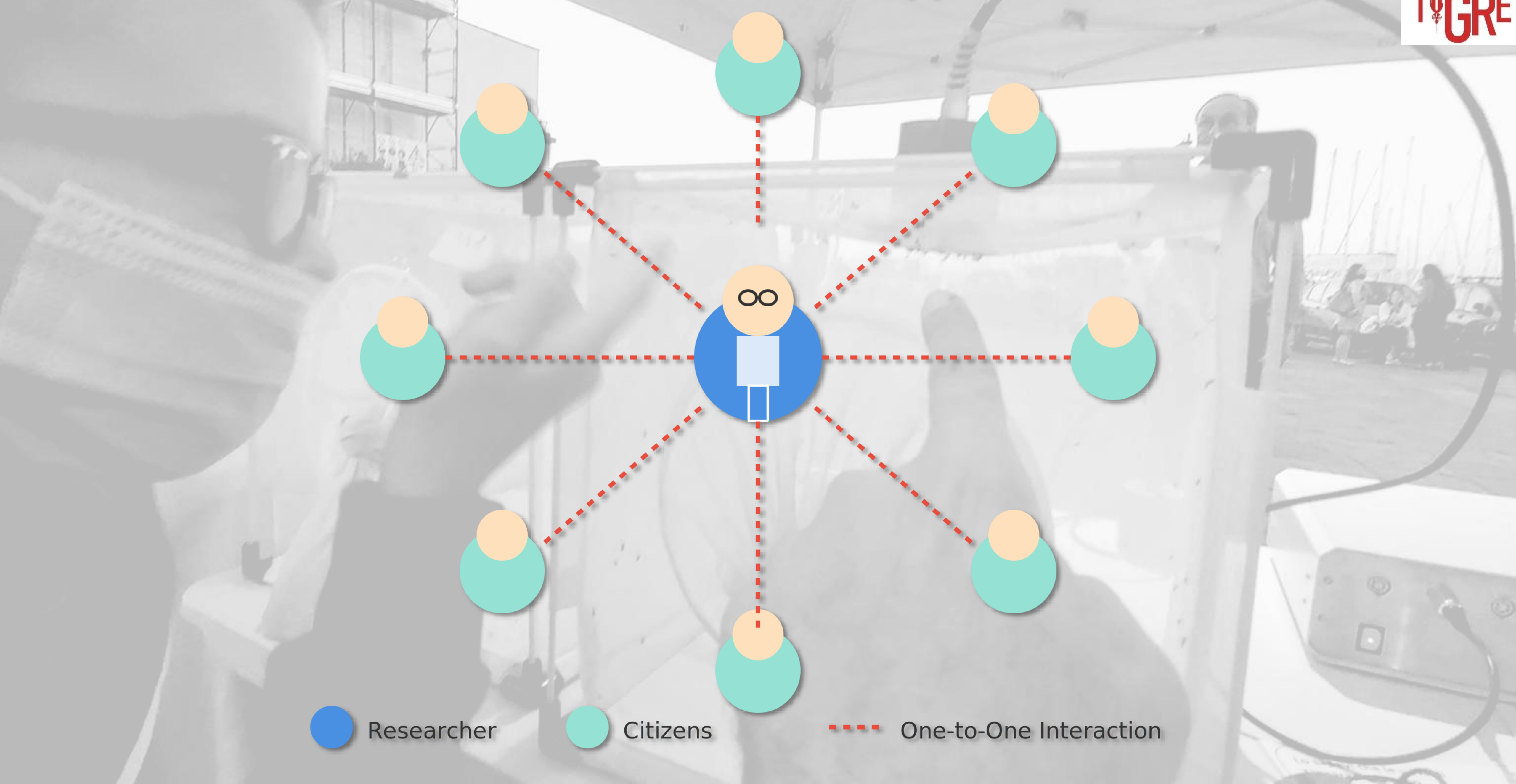
Abstract

In the last decades, the colonization of Mediterranean Europe and of other temperate regions by *Aedes albopictus* created an unprecedented nuisance problem in highly infested areas and new public health threats due to the vector competence of the species. The Ster-

MEETMETONIGHT

2018 – 2019 - 2020 – 2021

Researcher-Centered Approach



COMMUNITY ENGAGEMENT...

HOW to involve a whole community instead of hundreds of citizens???

2019

HOW WE MET

nuove tecnologie dell'arte

Course of New Technologies of Arts of the Fine Arts Academy of Naples

ABOUT US

nuove
tecnologi
dell'arte

STOP TIGRE

The French curator Nicolas Bourriaud published a book called *Relational Aesthetics* in 1998 in which he defined the term as:

> A set of artistic practices which take as their theoretical and practical point of departure the whole of human relations and their social context, rather than an independent and private space

He saw artists as facilitators rather than makers and regarded art as information exchanged between the artist and the viewers. The artist, in this sense, gives audiences access to power and the means to change the world.

In relational art, the audience is envisaged as a community. Rather than the artwork being an encounter between a viewer and an object, relational art produces encounters between people. Through these encounters, meaning is elaborated *collectively*, rather than in the space of individual consumption.

Bourriaud, Nicolas (2002). Relational Aesthetics. Pp. 17-18.

TATE

ART & ARTISTS EXHIBITIONS & EVENTS PLAN YOUR VISIT

ART TERM

RELATIONAL AESTHETICS

Term created by curator Nicolas Bourriaud in the 1990s to describe the tendency to make art based on, or inspired by, human relations and their social context

https://www.tate.org.uk/art/art-terms/r/relational-aesthetics

TIGER MOSQUITO

Monitoring and control campaign

A socio-cultural event focused on the fight against mosquitoes which, at the same time, could nourish the sense of **COMMUNITY** and of **CARE** for the territory.

nuove tecnologie dell'arte

Relational accelerator devices

(GIFT)

DONO

PHASE 1

PHASE 1

MURO

(WALL)

PHASE 2

PHASE 2

PHASE 2

MAY

3D CENSUS


PHASE 2

3D CENSUS OF PROCIDA POPULATION

nuove tecnologie dell'arte

HOME CHI SIAMO MANIFESTO PROGETTI BLOG PATTO FORMATIVO STATO DELL'ARTE CONTATTACI

PHASE 2


NOI
PHASE 2

(Artwork-EVENT)

OPERA evento

[A public event, co-organized with citizens, defined as "relational anthropological sculpture". A shared moment during which the relationships built during the project become the raw "material itself" by which the artwork-event is composed]

PHASE 3


98.

STOP TIGRE

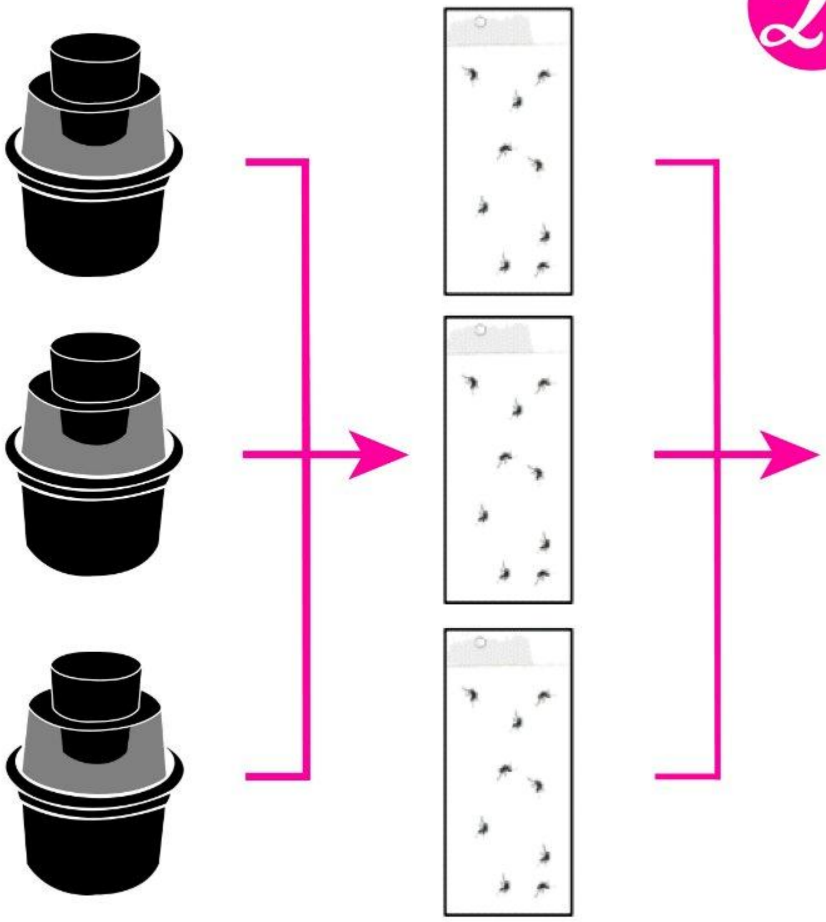
PHASE 2.1


ABBIAMO
POSIZIONATO
500
GRAVITRAPPOLE

PROCIDA
2022

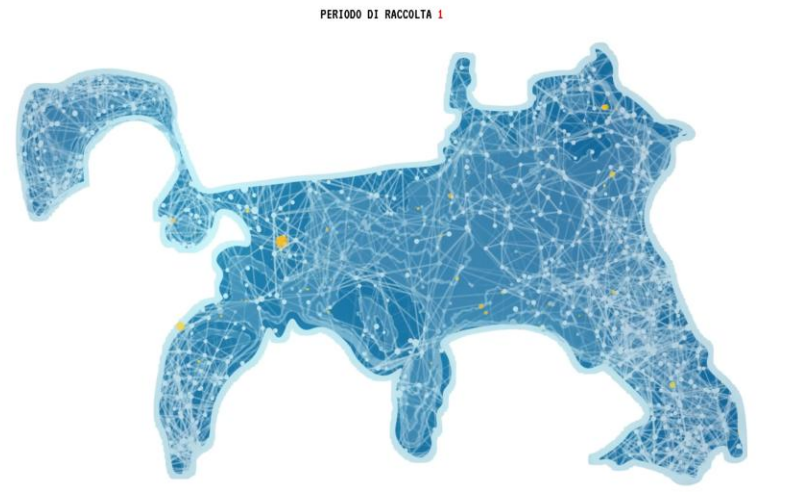
In just 5 days!!!

PHASE 2.1



2 PHOTO COLLECTION BY ZZZAPP

4 DYNAMIC DATA PRESENTATION BY NTA



3 AI MOSQUITO COUNTING AT DEPARTMENT OF BIOLOGY – FEDERICO II

1 CAPTURES AND PHOTOS BY CITIZENS

REAL-TIME DATA ANALYSIS AND VISUAL DATA PRESENTATION

STOP TIGRE NON MA io noi

2024

STOP TIGRE

PNRR
#NEXTGENERATIONITALIA

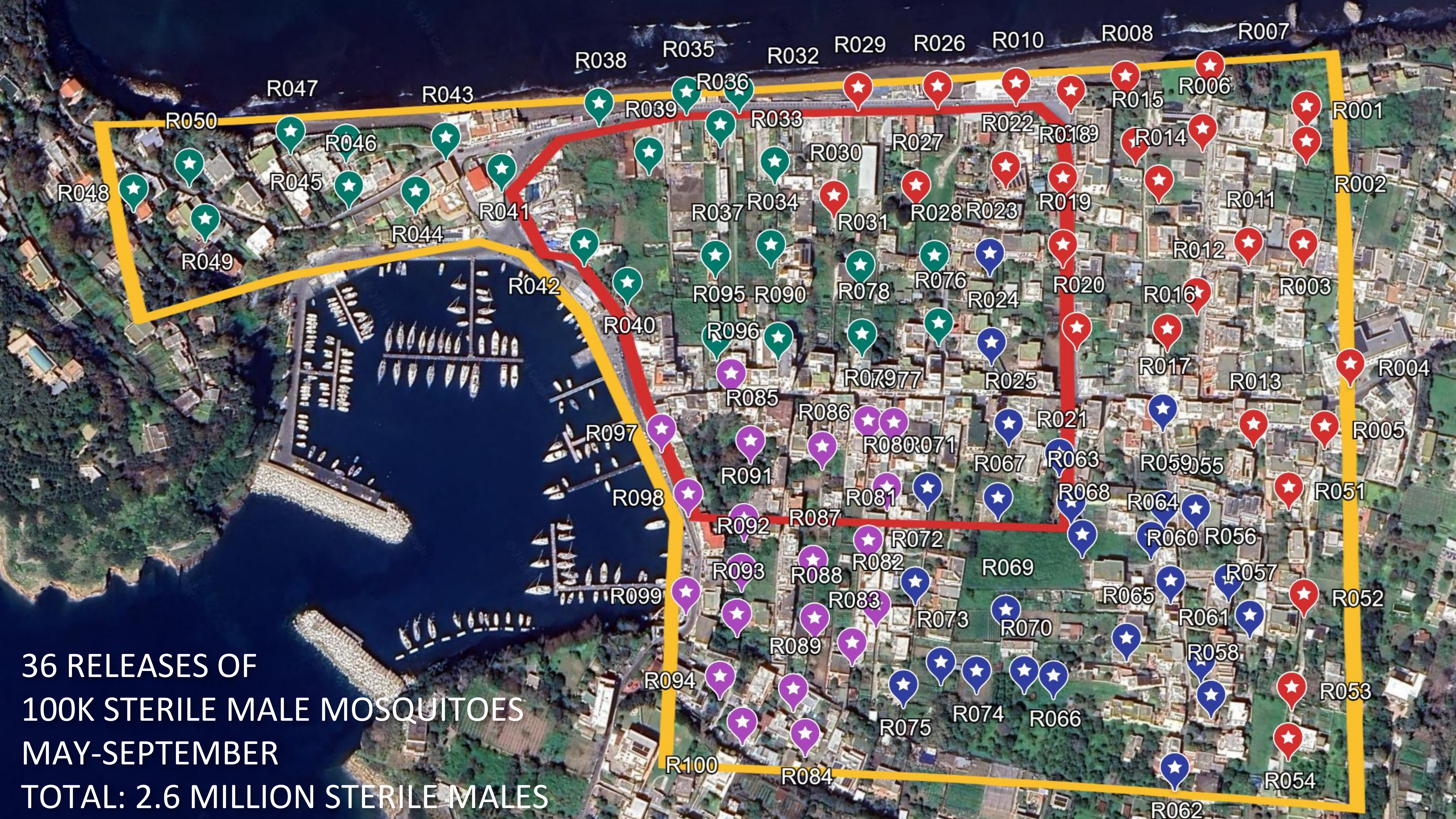
INF-ACT

nuove
tecnologie
dell'arte

NOI
UMANI
POWERED BY
ATTO D'AMORE

Finanziato
dall'Unione europea
NextGenerationEU


TEST AREA
CONTROL AREA


R038
R035
R032
R029
R026
R010
R008
R007
R047
R043
R036
R006
R050
R039
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R001
R046
R022
R014
R030
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R017
R004
R013
R085
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R086
R021
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R091
R067
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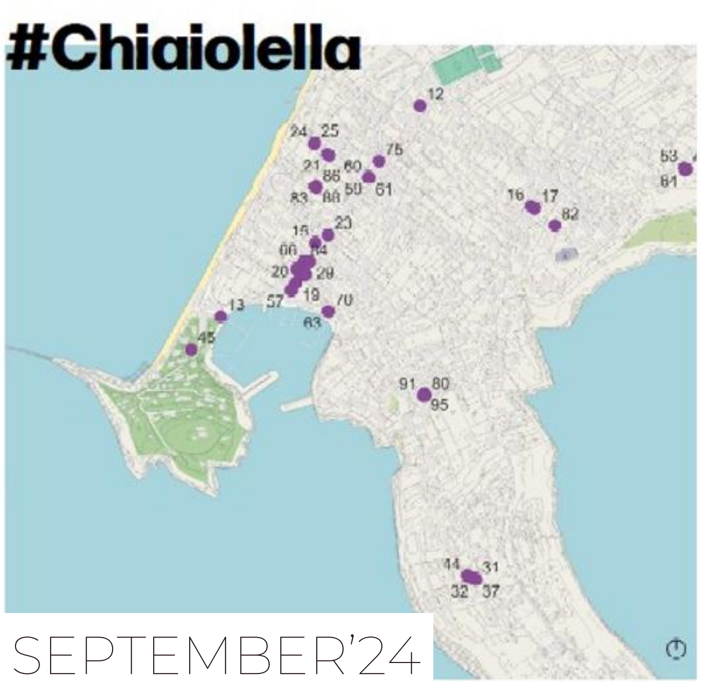
R051
R098
R081
R068
R064
R092
R087
R060
R056
R072
R082
R069
R093
R088
R057
R099
R065
R052
R083
R061
R073
R070
R058
R089
R094
R053
R074
R066
R075
R100
R084
R054
R062
36 RELEASES OF
100K STERILE MALE MOSQUITOES
MAY-SEPTEMBER
TOTAL: 2.6 MILLION STERILE MALES

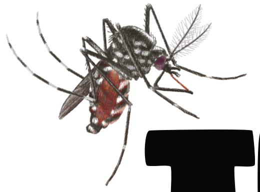


Citizen releases

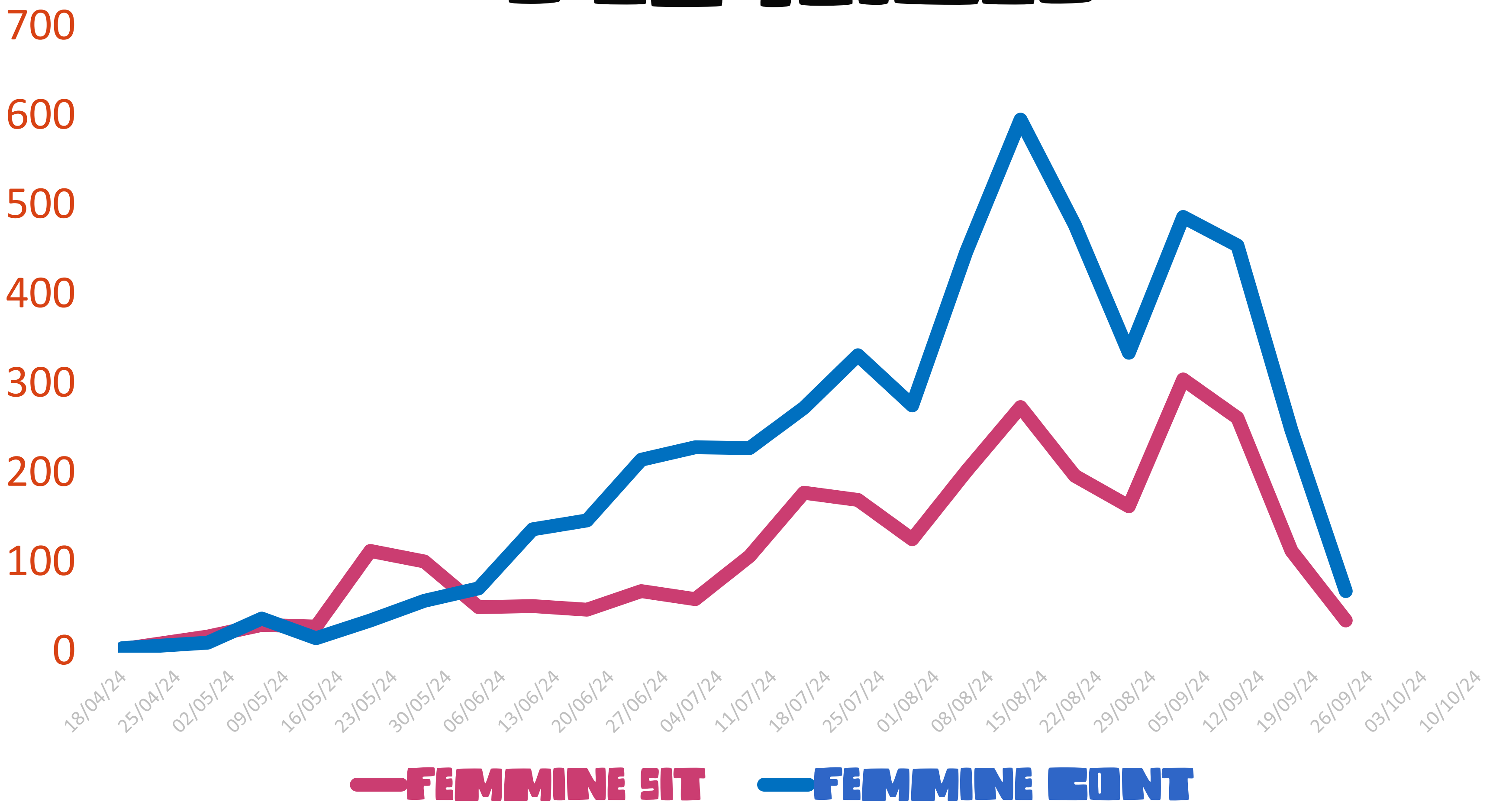
120,000 sterile males distributed over four months

89 families involved

IMPACT?



TOTAL GRAVID FEMALES



REDUCTION IN THE POPULATION OF PREGNANT FEMALES OF TIGER MOSQUITO

-55%

IN THE TEST AREA WITH RESPECT TO THE CONTROL AREA

(CAUGHT WITH GRAVID TRAPS)


WHAT'S NEXT?


TEST
ISLAND!!

The Procida case study

IL CASO PROCIDA

THINK GLOBAL, ACT LOCAL

Salvemini lab @ UNINA

nuove tecnologie dell'arte

THANK YOU!